

# AI-Powered Customer Sentiment Analysis for E-Commerce Websites Using Hybrid Machine Learning and Deep Learning Techniques

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## ABSTRACT

The project “AI-Powered Customer Sentiment Analysis for E-Commerce Websites” is designed to analyze customer reviews and classify them into positive, neutral, and negative sentiments using Artificial Intelligence. The system allows users to enter a product URL, from which real-time reviews are collected using web scraping techniques. These reviews are then preprocessed and converted into numerical features using TF-IDF for analysis.

The system includes an interactive frontend developed using HTML, CSS, and JavaScript, providing user-friendly pages such as login, registration, input forms, and dashboards with animations and visual insights. Multiple Machine Learning models such as Naive Bayes, SVM, Logistic Regression, Decision Tree, and KNN, along with Deep Learning models like CNN, LSTM, BiLSTM, and BERT, are trained and evaluated. A Hybrid model (CNN + SVM) is also used to improve performance, and the best model is saved for prediction.

The backend is implemented using Flask, where results are displayed on a dashboard with insights, trends, and sentiment distribution. Additional features include emotion detection, smart reply generation, and downloadable reports, with a MongoDB database used to store user data and analysis history. This system helps businesses understand customer feedback and make better decisions.

## CCS CONCEPTS

- Information systems → Data mining; Web applications
- Software and its engineering → Software architectures; Designing software systems

## KEYWORDS

Sentiment Analysis, Machine Learning, Deep Learning, TF-IDF, Hybrid Model, E-Commerce Analytics, Web Scraping,

**Lay Abstract:** Sentiment analysis is an important application of Artificial Intelligence that helps in understanding customer opinions from textual data such as product reviews [1]. With the rapid growth of e-commerce platforms like Flipkart and Walmart, a massive amount of customer feedback is generated daily. Analyzing this data manually is difficult, time-consuming, and inefficient.

This project presents an AI-powered system that automatically collects customer reviews from e-commerce websites and

analyzes them to determine whether the sentiment is positive, negative, or neutral. The system uses Machine Learning and Deep Learning models to understand patterns in text and provide accurate predictions. It also includes a hybrid model combining Convolutional Neural Networks (CNN) and Support Vector Machines (SVM) to improve performance.

The system processes data in multiple stages, including web scraping, text preprocessing, feature extraction using TF-IDF, and sentiment classification. The results are displayed through an interactive dashboard that shows sentiment distribution, average rating, and top reviews. Additional features such as emotion detection and smart reply generation enhance the usability of the system.

This solution helps businesses quickly understand customer opinions, identify product strengths and weaknesses, and make better decisions. By automating sentiment analysis and providing real-time insights, the system improves efficiency and supports data-driven strategies in e-commerce.

## 1. INTRODUCTION

The rapid growth of digital technologies and internet penetration has significantly transformed the global retail landscape, leading to the exponential rise of e-commerce platforms such as Flipkart and Walmart. These platforms generate massive volumes of user-generated content in the form of product reviews, ratings, feedback comments, and discussion threads. Customer reviews have become a critical component in shaping purchasing decisions, influencing brand reputation, and guiding product development strategies. In today's highly competitive digital marketplace, understanding customer sentiment is not merely beneficial but essential for business growth and sustainability.

With millions of reviews being posted daily, manual analysis of customer feedback is impractical, time-consuming, and prone to human bias. Customers express their opinions in diverse forms—ranging from structured ratings to unstructured textual reviews that often include slang, abbreviations, emotional expressions, and contextual nuances. This complexity introduces significant challenges for traditional data processing systems that rely on structured inputs or simple keyword-based approaches.

Consequently, there is an increasing demand for intelligent systems capable of automatically analyzing and interpreting customer sentiments with high accuracy and efficiency.

Artificial Intelligence (AI), combined with advancements in

Natural Language Processing (NLP)-enabled Machine Learning (ML), has emerged as a powerful solution to this problem. Sentiment analysis, also known as opinion mining, is a subfield of NLP that focuses on identifying and classifying opinions expressed in textual data. It enables systems to determine whether a piece of text conveys a positive, negative, or neutral sentiment. By leveraging AI-powered sentiment analysis, businesses can extract valuable insights from customer feedback, identify product strengths and weaknesses, monitor brand perception, and improve overall customer experience [1].

The project titled “**AI-Powered Customer Sentiment Analysis for E-Commerce Websites**” is designed to address these challenges by developing an intelligent, scalable, and automated system for analyzing customer reviews. The proposed system allows users to input product URLs from popular e-commerce platforms like Flipkart and Walmart. Using web scraping techniques, the system collects real-time customer reviews directly from these platforms. This capability ensures that the analysis is always based on up-to-date data, making it highly relevant for dynamic market environments.

Once the reviews are collected, they undergo a series of preprocessing steps to prepare the text for analysis. These steps include removal of noise such as punctuation and special characters, conversion of text to lowercase, tokenization, stop-word removal, and normalization. Preprocessing is a crucial phase as it improves data quality and enhances the performance of machine learning models. After preprocessing, the textual data is transformed into numerical representations using Term Frequency-Inverse Document Frequency (TF-IDF), a widely used feature extraction technique in NLP. TF-IDF helps in identifying important words in a document by considering both their frequency and their uniqueness across the dataset.

The core strength of the proposed system lies in its hybrid modeling approach. Instead of relying on a single algorithm, the system integrates multiple machine learning and deep learning models to achieve better performance. Traditional machine learning models such as Naive Bayes, Support Vector Machine (SVM) [7], Logistic Regression, Decision Tree, and K-Nearest Neighbors (KNN) are implemented due to their efficiency in handling structured numerical data and their proven effectiveness in classification tasks. These models provide a strong baseline for sentiment classification.

In addition to traditional models, the system incorporates advanced deep learning techniques including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) [6], Bidirectional LSTM (BiLSTM), and transformer-based models such as BERT (Bidirectional Encoder Representations from Transformers). Deep learning models are particularly effective in capturing contextual and sequential information in text data, allowing for a deeper understanding of language patterns and sentiment expressions. Among these, BERT stands out due to its ability to process text bidirectionally, enabling it to understand context more effectively than earlier models.

To further enhance system performance, a hybrid model combining CNN and SVM is developed. This approach leverages the feature extraction capabilities of CNN and the classification strength of SVM, resulting in improved accuracy and robustness. The system evaluates all models using

appropriate performance metrics and selects the best-performing model for final deployment. This model is then saved and used for real-time sentiment prediction.

The system architecture is designed using a client-server model, where the frontend provides an interactive user interface and the backend handles data processing and model execution. The frontend is developed using HTML, CSS, and JavaScript, offering user-friendly features such as login and registration pages, input forms for URL submission, and dashboards displaying sentiment analysis results. The dashboard includes visualizations such as charts, graphs, and sentiment distribution plots, making it easier for users to interpret the results.

The backend is implemented using the Flask framework, which acts as a bridge between the user interface and the machine learning models. Flask handles API requests, processes user inputs, performs sentiment analysis, and sends the results back to the frontend. The system also integrates a MongoDB database to store user information, analysis history, and generated reports. This enables users to access past analyses and track sentiment trends over time.

Beyond basic sentiment classification, the system includes advanced features such as emotion detection, smart reply generation, and report generation. Emotion detection identifies specific emotional tones such as happiness, anger, or frustration in customer reviews, providing deeper insights into customer behavior. Smart reply generation suggests appropriate responses to customer feedback, which can be useful for customer support teams. The system also allows users to download analysis reports, making it suitable for business applications and decision-making processes.

The significance of this research lies in its ability to bridge the gap between raw customer feedback and actionable business insights. By automating the process of sentiment analysis, the system reduces manual effort, improves accuracy, and enables real-time decision-making. It provides businesses with a powerful tool to monitor customer satisfaction, identify emerging issues, and respond proactively to customer needs.

Furthermore, the system is designed to be scalable and adaptable, allowing it to handle large volumes of data and integrate new models or features in the future. Its modular architecture ensures that components can be updated or replaced without affecting the overall system functionality. This makes it suitable for deployment in real-world e-commerce environments where data volume and complexity are continuously increasing.

In the broader context, this research contributes to the field of artificial intelligence by demonstrating the practical application of machine learning and deep learning techniques [12] in real-world scenarios. It highlights the importance of hybrid models in improving classification performance and showcases the potential of AI-driven systems in transforming business operations. The integration of web scraping, NLP, machine learning, and web technologies into a single system represents a comprehensive approach to solving complex data analysis problems.

In conclusion, the proposed AI-powered customer sentiment analysis system provides an efficient, accurate, and scalable solution for analyzing e-commerce reviews. By leveraging advanced AI techniques and modern web technologies, it enables businesses to gain valuable insights from customer feedback and make informed decisions. As e-commerce continues to grow, such intelligent systems will play a crucial role in enhancing customer

## 2. BACKGROUND AND RELATED WORK

We first provide an overview of important concepts such as ML-enabled systems, sentiment analysis in e-commerce, and hybrid learning approaches, and then discuss related work in this domain.

### 2.1 ML-Enabled Systems in Sentiment Analysis

Machine Learning (ML) is a subfield of Artificial Intelligence that focuses on developing algorithms capable of learning patterns from data without being explicitly programmed. ML models are trained using historical data to identify relationships and patterns, which can then be used to make predictions on unseen data. In the context of sentiment analysis, ML models learn linguistic patterns and word associations to classify textual data into predefined categories such as positive, negative, and neutral sentiments.

According to standard ML development practices, building an ML model involves several stages, including data collection, preprocessing, feature extraction, model training, evaluation, deployment, and monitoring [2]. These stages often involve iterative feedback loops to improve performance and accuracy.

In this project, the sentiment analysis system is an example of an ML-enabled system, as it integrates trained machine learning and deep learning models within a software application. The system processes customer reviews collected from e-commerce platforms such as Flipkart and Walmart, and generates predictions based on learned patterns.

However, ML-enabled systems introduce several challenges, including:

- Dependency on training data quality
- Lack of transparency in model decisions
- Variability in prediction accuracy
- Difficulty in handling unstructured text

To address these challenges, the proposed system incorporates multiple models and uses a hybrid approach to improve reliability and performance.

### 2.2 Text Processing and Sentiment Analysis Techniques

Sentiment analysis is a Natural Language Processing (NLP) technique used to extract subjective information from textual data. In e-commerce applications, it plays a critical role in understanding customer feedback and improving product quality. Textual data in reviews is often unstructured and contains noise such as punctuation, slang, and irrelevant words. Therefore, preprocessing is a crucial step in sentiment analysis. It includes operations such as text cleaning, tokenization, stop-word removal, and normalization.

After preprocessing, textual data must be converted into numerical form for machine learning models. One of the most widely used techniques is TF-IDF (Term Frequency–Inverse Document Frequency), which measures the importance of a word in a document relative to a collection of documents [10]. TF-IDF assigns higher weights to words that are unique and informative, while reducing the impact of commonly occurring words.

In this system, TF-IDF is used as the primary feature extraction method. The processed data is then fed into classification models such as Logistic Regression, Naive Bayes [9], and Support Vector Machines. Additionally, deep learning models such as LSTM and BERT are used to capture contextual relationships in text data.

### 2.3 Hybrid Learning Approaches and System Design

Traditional machine learning models are efficient for classification tasks but may fail to capture complex linguistic patterns. Deep learning models, on the other hand, are capable of understanding context and sequence but require significant computational resources. Hybrid approaches combine the strengths of both methods to achieve better performance.

In this project, a hybrid model combining Convolutional Neural Networks (CNN) [5] and Support Vector Machines (SVM) is used. The CNN model acts as a feature extractor, identifying important patterns in text, while the SVM model performs classification based on these features. This combination improves accuracy and robustness compared to individual models.

The system architecture is designed to support real-time sentiment analysis. It includes:

- A frontend interface for user interaction
- A backend server implemented using Flask
- A machine learning layer for prediction
- A database system for storing results

The system follows a pipeline-based architecture, where data flows through multiple stages, including web scraping, preprocessing, feature extraction, prediction, and visualization.

## 3 STUDY DESIGN AND EXECUTION

he study of execution focuses on how the proposed AI-powered sentiment analysis system operates in a real-world environment. It explains the internal workflow, data flow, model execution, and system behavior during runtime. Unlike the design phase, which defines system structure, the execution phase emphasizes how different components interact dynamically to process input data and generate meaningful outputs.

The proposed system is designed to handle real-time data processing, ensuring efficient and accurate sentiment prediction for large volumes of customer reviews. It integrates multiple technologies, including web scraping, natural language processing, machine learning, and data visualization, into a unified pipeline.

### 3.1 Execution Environment

The system is executed in a client-server architecture where the frontend interacts with the backend server, which in turn communicates with the machine learning models and database.

The execution environment includes:

- Programming Language: Python
- Framework: Flask
- Machine Learning Libraries: Scikit-learn [11], TensorFlow/Keras
- Database: MongoDB [15]
- Web Technologies: HTML, CSS, JavaScript

The system is designed to run on standard computing hardware,

Analysis of Feasibility and Tactics for ML-Enabled Systems without requiring specialized infrastructure.

### 3.2 Runtime Data Flow

During execution, the system follows a structured pipeline that processes data in sequential stages. Each stage performs a specific transformation on the data, ensuring smooth flow from input to output.

#### Step 1: User Input Handling

The execution begins when a user enters a product URL into the frontend interface. This input is sent to the backend server via an HTTP request.

#### Step 2: Real-Time Data Extraction

The backend triggers the web scraping module, which retrieves customer reviews from e-commerce platforms. This process involves:

- Sending HTTP requests to fetch webpage content
- Parsing HTML using scraping libraries
- Extracting relevant review text and ratings

The system ensures that only valid and meaningful review data is collected.

### 3.3 Text Processing Execution

Text preprocessing is a critical step in execution as raw data often contains noise and inconsistencies.

#### Cleaning Process

- Conversion to lowercase
- Removal of special characters and numbers
- Elimination of extra spaces

#### Normalization

- Standardizing words
- Removing irrelevant tokens

This ensures uniformity in input data, improving model performance.

### 3.4 Feature Extraction Execution

After preprocessing, the system converts textual data into numerical form using TF-IDF (Term Frequency–Inverse Document Frequency).

#### Execution Mechanism

- Each review is transformed into a vector representation
- Words are assigned weights based on importance
- Rare but meaningful words receive higher weights

This transformation allows machine learning models to interpret textual information mathematically.

### 3.5 Model Execution and Prediction

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The prediction phase is the core execution stage where the system classifies sentiments.

#### Model Loading

- Pre-trained models are loaded from .pkl files
- Vectorizer and classifier are initialized
- Validation checks ensure models are properly fitted

#### Prediction Process

- Input vector is passed to the model
- The model applies learned patterns
- Output is generated as:
  - Positive
  - Negative
  - Neutral

#### Hybrid Model Execution

In the hybrid CNN-SVM model:

- CNN extracts deep features from text
- SVM performs final classification
- This combination improves accuracy and stability

### 3.6 Aggregation and Rating Calculation

Once individual predictions are generated, the system aggregates results to provide overall insights.

#### Sentiment Distribution

- Percentage of positive, negative, and neutral reviews
- Visualization using pie charts

#### Rating Calculation

The average rating is computed using:

- Mapping sentiments to scores:
  - Positive = 5
  - Neutral = 3
  - Negative = 1
- Calculating weighted average:

$$\text{Average Rating} = \frac{\sum(\text{Sentiment Score} \times \text{Count})}{\text{Total Reviews}}$$

This provides a numerical representation of overall product performance.

### 3.7 Smart Reply Generation System

The Smart Reply Generation System is an intelligent module designed to automatically generate appropriate responses to customer reviews based on their sentiment and context. In modern

CC-0. Since April 2024, handling large volumes of customer feedback manually is both time-consuming and inefficient. This module addresses that challenge by providing automated, consistent, and context-aware replies, thereby improving the efficiency of customer support operations.

From a theoretical perspective, the system is based on Natural Language Processing (NLP) techniques and sentiment classification. Once the sentiment of a review is determined—whether positive, negative, or neutral—the system uses predefined linguistic patterns and response templates to generate a suitable reply. The logic behind this approach is that customer responses can be categorized based on sentiment polarity, allowing the system to map each category to a corresponding response style. While more advanced systems may use deep learning models such as sequence-to-sequence architectures or transformer-based models for dynamic text generation, this implementation adopts a template-based approach to ensure reliability, simplicity, and faster execution.

In terms of design, the Smart Reply module is integrated as a post-processing component within the sentiment analysis pipeline. After the prediction stage, the system passes the classified sentiment along with the original review text to the reply generation unit. This unit selects an appropriate response template and, in some cases, customizes it using keywords extracted from the review. For example, if a review is classified as negative, the system generates a response expressing apology and assurance of improvement. Similarly, positive reviews are acknowledged with gratitude, and neutral reviews receive balanced and polite responses.

During execution, the system processes each review individually. After sentiment prediction, a corresponding reply is generated and displayed in the dashboard under a dedicated section such as “Draft Reply.” This enables customer support teams to either directly use the generated response or modify it slightly before sending it to the customer. The automation of this process significantly reduces response time and ensures uniform communication standards across the platform.

The Smart Reply System offers several advantages, including improved efficiency, reduced manual workload, and enhanced customer satisfaction through timely responses. However, it also has certain limitations. Since the system relies on predefined templates, it may lack deep personalization and may not fully capture complex linguistic nuances such as sarcasm or mixed sentiments. Despite these limitations, the module serves as a practical and effective solution for large-scale customer interaction management.

### 3.8 Emotion Detection Module

The Emotion Detection Module extends the capabilities of traditional sentiment analysis by identifying specific emotional states expressed in customer reviews. While sentiment analysis classifies text into broad categories such as positive, negative, or neutral, emotion detection provides a more fine-grained understanding of user feelings, including emotions such as happiness, anger, sadness, and neutrality. This deeper level of analysis enables businesses to better understand customer behavior and respond more effectively to feedback.

The theoretical foundation of this module lies in advanced Natural Language Processing techniques that analyze linguistic

patterns, word usage, and contextual relationships to infer emotional states. Unlike sentiment analysis, which focuses primarily on polarity, emotion detection captures the intensity and nature of the expressed sentiment. Machine learning and deep learning models, particularly those based on sequential architectures like LSTM or transformer-based models like BERT [4], are well-suited for this task due to their ability to understand context and long-range dependencies in text.

From a design perspective, the Emotion Detection Module operates in parallel with the sentiment analysis system. After preprocessing, the cleaned textual data is passed through a feature extraction stage, typically using TF-IDF or word embeddings. The resulting feature vectors are then input into a trained classification model that predicts the most likely emotional category. The module is designed to classify emotions into four primary categories: happy, angry, sad, and neutral, although the architecture allows for future expansion to include additional emotional states.

During execution, each review is processed through the emotion detection pipeline after or alongside sentiment prediction. The model evaluates the input text and assigns probabilities to each emotion class. The emotion with the highest probability is selected as the final output and displayed on the dashboard along with the sentiment classification. This combined output provides a more comprehensive view of customer feedback, enabling better analysis and interpretation.

The importance of emotion detection lies in its ability to reveal deeper insights into customer experiences. For instance, identifying anger in a review may indicate a critical issue requiring immediate attention, while detecting happiness may highlight aspects of the product that resonate positively with users. This information can be used to improve product quality, enhance customer service strategies, and support data-driven decision-making.

Despite its advantages, the module also faces certain challenges. Detecting emotions accurately can be difficult in cases of sarcasm, ambiguous language, or mixed emotions within a single review. Additionally, the performance of the model heavily depends on the quality and diversity of the training dataset. Nevertheless, the integration of emotion detection significantly enhances the analytical capabilities of the system and provides valuable insights beyond traditional sentiment classification.

## 4 RESULTS

The proposed AI-Powered Customer Sentiment Analysis for E-Commerce Websites was successfully implemented and evaluated using real-world customer reviews collected from online shopping platforms such as Flipkart and Walmart. The developed system was able to automatically extract customer reviews, preprocess the textual data, classify sentiments, detect emotions, generate ratings, and present the final outcomes through an interactive dashboard. The results demonstrate that the proposed hybrid machine learning framework is effective for analyzing large volumes of customer feedback in real time.

Initially, multiple machine learning and deep learning models were trained and tested using labeled review datasets. Traditional machine learning models such as Naive Bayes, Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN) were evaluated for baseline



performance. In addition, traditional models like Naïve Bayes, CNN, LSTM, BiLSTM, and BERT were implemented to capture complex contextual information from text data. After comparative analysis, a Hybrid CNN + SVM model produced superior results in terms of classification accuracy, robustness, and generalization performance.

The final system is organized into several result-oriented modules including review collection, preprocessing, feature extraction, sentiment prediction, emotion detection, smart reply generation, rating prediction, and visualization. These modules operate sequentially in a pipeline architecture, allowing efficient processing of real-time data. The integration of all modules ensures that the system not only predicts sentiment labels but also provides actionable business insights.

The experimental results indicate that preprocessing techniques such as lowercasing, punctuation removal, stop-word elimination, and stemming significantly improved the quality of predictions. Similarly, TF-IDF feature extraction enhanced the representation of textual data by converting words into weighted numerical vectors. These vectors were then used as input for classification models.

The dashboard results showed clear sentiment distributions across products. Positive reviews dominated highly rated products, while negative reviews highlighted issues related to quality, delivery delays, and damaged items. Neutral reviews generally represented balanced or mixed opinions. Emotion detection further classified reviews into Happy, Angry, Sad, and Neutral categories, providing deeper insight beyond basic sentiment analysis.

The Smart Reply Generation module successfully generated automated responses based on predicted sentiments. Positive reviews received appreciation messages, while negative reviews generated apology and support-oriented responses. This feature demonstrated practical value for automating customer engagement.

The Rating Prediction module converted review sentiments into star-based ratings ranging from 1 to 5. Products with a higher proportion of positive reviews received better overall ratings, while those with frequent complaints obtained lower scores. This helps users quickly understand product quality through aggregated sentiment-based ratings.

The proposed system also demonstrated strong scalability and real-time applicability. Reviews scraped from live product URLs were analyzed within a short response time, proving the feasibility of deploying the system for real-world e-commerce environments.

4.1 Comparative Performance of Models

A major part of the results section involved comparing different machine learning and deep learning models. Each model was evaluated using performance metrics such as Accuracy, Precision, Recall, and F1-score. Accuracy measured the overall correctness of predictions, Precision measured the proportion of correct positive predictions, Recall measured how many actual positives were identified, and F1-score provided a balanced evaluation of Precision and Recall.

Traditional machine learning models performed well on structured review datasets. Naive Bayes was fast and

lightweight, making it useful for quick predictions on large datasets. Logistic Regression provided stable performance and was effective for linearly separable data. Decision Trees offered easy interpretability but were more prone to overfitting. KNN worked adequately but became slower for large datasets because it compares new samples with many training examples.

Support Vector Machine (SVM) delivered strong performance because it effectively separates sentiment classes using optimal decision boundaries. Deep learning models further improved results by capturing word order, context, and semantic relationships. CNN identified important local patterns such as frequent opinion phrases. LSTM learned long-term dependencies in review sentences, while BiLSTM processed sequences in both forward and backward directions for better contextual understanding. BERT produced advanced contextual embeddings and achieved very high accuracy.

The Hybrid CNN + SVM model outperformed all other approaches. CNN acted as an automatic feature extractor, while SVM used these learned features for final sentiment classification. This combination reduced feature engineering effort and increased robustness.

These results confirm that hybrid architectures can combine the strengths of multiple learning methods for improved performance. Overall, the experimental results indicate that hybrid approaches provide a balanced combination of accuracy, efficiency, and scalability for real-time e-commerce sentiment analysis systems.

| Model               | Accuracy (%) | Precision | Recall | F1 Score |
|---------------------|--------------|-----------|--------|----------|
| Naive Bayes         | 82           | 0.81      | 0.80   | 0.80     |
| Logistic Regression | 88           | 0.87      | 0.88   | 0.87     |
| SVM                 | 91           | 0.90      | 0.91   | 0.90     |
| Decision Tree       | 80           | 0.79      | 0.78   | 0.78     |
| KNN                 | 78           | 0.77      | 0.76   | 0.76     |
| CNN                 | 92           | 0.91      | 0.92   | 0.91     |
| LSTM                | 93           | 0.92      | 0.93   | 0.92     |
| BiLSTM              | 94           | 0.93      | 0.94   | 0.93     |
| BERT                | 95           | 0.94      | 0.95   | 0.94     |
| Hybrid CNN+SVM      | 96           | 0.95      | 0.96   | 0.95     |

4.2 Sentiment Distribution Results

The system analyzed large volumes of reviews and classified them into Positive, Neutral, and Negative categories. Most products with good quality and timely delivery showed higher positive sentiment percentages. Products with defects, poor packaging, or delayed shipping showed higher negative sentiment values.

| Sentiment Class | Percentage |
|-----------------|------------|
| Positive        | 64%        |
| Neutral         | 18%        |
| Negative        | 18%        |

The predominance of positive sentiment indicates that many

customers were 14 and 20, with the top purchases. Neutral reviews often contained balanced opinions or short factual comments. Negative reviews highlighted valuable improvement areas such as damaged products, misleading descriptions, refund issues, or poor service. This classification allows businesses to quickly understand product reputation and identify critical customer complaints.

### 4.3 Emotion Detection Results

Beyond sentiment analysis, the system successfully detected emotions hidden inside customer reviews. This is important because two negative reviews may express different emotional intensity. One may express sadness, while another may show anger.

| Emotion | Percentage |
|---------|------------|
| Happy   | 52%        |
| Angry   | 16%        |
| Sad     | 12%        |
| Neutral | 20%        |

Happy emotions were mainly associated with product satisfaction, good pricing, or quick delivery. Angry emotions were linked with refund issues, damaged items, or poor customer support. Sad emotions reflected disappointment or unmet expectations. Neutral emotions represented objective descriptions without strong emotional signals.

Emotion detection provides deeper psychological insight into customer behavior and helps companies prioritize severe complaints.

### 4.4 Rating Prediction Results

The project also included a sentiment-to-rating conversion mechanism. Instead of relying only on manually submitted ratings, the system estimated star ratings from textual reviews. This is especially useful when customers write reviews without giving star scores.

| Sentiment Score | Predicted Rating |
|-----------------|------------------|
| Highly Positive | 5 Star           |
| Positive        | 4 Star           |
| Neutral         | 3 Star           |
| Negative        | 2 Star           |
| Highly Negative | 1 Star           |

For example, a review such as “Excellent product, highly recommended” would be predicted as 5 stars, while a review such as “Very poor quality, waste of money” would be predicted as 1 star. This module improves product evaluation systems by generating ratings from review text automatically.

### 4.5 Smart Reply Generation Results

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The Smart Reply module successfully generated automated responses according to sentiment class. This helps customer support teams save time and maintain consistent communication.

Positive reviews generated appreciation responses such as thanking the customer for valuable feedback. Negative reviews generated apology-oriented responses with assurance of resolution. Neutral reviews received balanced professional responses.

Examples include:

Positive Review: “Thank you for your valuable feedback. We are glad you enjoyed the product.”

Negative Review: “We apologize for the inconvenience caused. We will improve our service.”

Neutral Review: “Thank you for sharing your experience with us.”

This feature can significantly reduce manual workload in large e-commerce environments.

### 4.6 Dashboard Visualization Results

The developed dashboard displayed all outputs in an easy-to-understand visual format. Pie charts showed sentiment distribution. Bar graphs compared ratings. Line graphs showed trends over time. Tables displayed top reviews and common complaint categories.

The dashboard allowed users to interactively inspect product feedback and make faster decisions. Business managers could instantly identify whether a product launch was successful or problematic. Visualization improved interpretability of machine learning results for non-technical users.

### 4.7 System Efficiency Results

The performance of the system was also measured in terms of execution speed and scalability. These results indicate that the system is capable of handling real-time business scenarios. Fast response time ensures practical usability.

| Parameter                   | Result       |
|-----------------------------|--------------|
| Average Prediction Time     | < 2 seconds  |
| URL Review Scraping Time    | 5–10 seconds |
| Dashboard Loading Time      | < 3 seconds  |
| Reviews Processed per Batch | 1000+        |

### 4.8 Practical Business Impact

The final results demonstrate that the project is not limited to academic sentiment classification but has direct industrial relevance. Businesses can use the system to monitor customer satisfaction, identify product weaknesses, improve support quality, optimize marketing campaigns, and increase customer retention.

Instead of manually reading thousands of reviews, decision-makers can rely on AI-generated summaries and analytics.

#### 4.9 Overall Result Summary

The developed AI-powered customer sentiment analysis system successfully integrates review scraping, preprocessing, feature extraction, sentiment classification, emotion detection, rating prediction, smart replies, and visualization into a single platform. The Hybrid CNN + SVM model achieved the highest predictive performance. The dashboard converted raw customer opinions into valuable business intelligence.

Therefore, the proposed system is accurate, scalable, efficient, and highly suitable for real-world e-commerce platforms such as Flipkart and Walmart.

## 5 Discussion

The proposed AI-powered customer sentiment analysis system demonstrates an effective integration of machine learning, deep learning, and natural language processing techniques for analyzing large volumes of customer feedback in e-commerce platforms such as Flipkart and Walmart. The system successfully processes unstructured textual data and converts it into meaningful insights, enabling businesses to better understand customer opinions and improve decision-making processes.

One of the key observations from the system's implementation is the effectiveness of combining traditional machine learning models with deep learning approaches. While models such as Logistic Regression and Support Vector Machines provide fast and efficient classification, deep learning models such as CNN and LSTM contribute to capturing contextual and sequential patterns in textual data. The hybrid CNN-SVM model, in particular, improves classification accuracy and robustness, highlighting the importance of integrating multiple learning techniques in sentiment analysis tasks.

Another important aspect of the system is its real-time data processing capability. By incorporating web scraping techniques, the system dynamically collects customer reviews directly from e-commerce platforms. This allows the analysis

to reflect current customer opinions rather than relying on static datasets. However, this approach also introduces challenges such as dependency on website structure, potential scraping restrictions, and variability in data quality, which can affect system performance.

The inclusion of advanced modules such as emotion detection and smart reply generation enhances the overall functionality of the system. Emotion detection provides deeper insights into customer behavior by identifying specific emotional states, while the smart reply system improves customer engagement by generating automated responses. These features demonstrate how sentiment analysis can be extended beyond basic classification to support real-world business applications.

Despite its advantages, the system has certain limitations. The accuracy of predictions depends heavily on the quality and size of the training dataset. Additionally, the system may struggle to correctly interpret sarcasm, slang, or mixed sentiments within a single review. Deep learning models also require higher computational resources, which can impact performance in resource-constrained environments. Furthermore, template-based smart replies may lack personalization, reducing their

effectiveness in complex scenarios.

Overall, the system provides a scalable and efficient solution for sentiment analysis in e-commerce applications. It highlights the potential of AI-driven techniques in transforming raw customer feedback into actionable insights. Future improvements can focus on enhancing model accuracy, incorporating multilingual support, and using advanced transformer-based architectures to further improve contextual understanding and system performance.

## 6 Future Work

The proposed AI-powered customer sentiment analysis system provides a strong foundation for analyzing customer feedback in e-commerce platforms, but there are several opportunities for further enhancement and expansion. One important area for future improvement is the integration of advanced transformer-based models such as BERT and its variants. These models are capable of capturing deeper contextual relationships within text, which can significantly improve the accuracy of sentiment and emotion classification, especially in complex linguistic scenarios such as sarcasm or mixed sentiments.

Another potential enhancement is the incorporation of multilingual sentiment analysis. Currently, the system primarily processes English-language reviews, which limits its applicability in global markets. Extending the system to support multiple languages would enable businesses to analyze customer feedback from diverse regions, thereby increasing its practical usability in international e-commerce platforms.

The system can also be improved by integrating real-time APIs instead of relying solely on web scraping techniques. While web scraping provides dynamic data collection, it is dependent on website structures and may face restrictions. API-based data retrieval would ensure more reliable, structured, and scalable access to customer reviews. Additionally, implementing streaming data processing techniques can further enhance real-time analytics capabilities.

Another promising direction is the use of more sophisticated text generation techniques for the Smart Reply System. Instead of

relying on predefined templates, advanced Natural Language Generation (NLG) models can be used to generate highly personalized and context-aware responses. This would improve customer engagement and provide a more human-like interaction experience.

The system can also be extended by incorporating recommendation systems that suggest product improvements or alternative products based on sentiment trends. Integration with business intelligence tools and dashboards can further enhance decision-making by providing predictive analytics and trend forecasting.

Finally, improvements in system performance and scalability can be achieved by deploying the application on cloud platforms and using distributed computing frameworks. This would allow the system to handle large-scale data efficiently and support real-world industrial applications.

## 7 Conclusion

In this research, an AI-powered customer sentiment analysis system for e-commerce websites has been successfully designed



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The proposed system integrates machine learning, deep learning, and natural language processing techniques to analyze customer reviews and classify them into positive, negative, and neutral sentiments. By integrating models such as Logistic Regression, Support Vector Machines, CNN, and LSTM, along with a hybrid CNN-SVM approach, the system achieves improved accuracy and robustness in sentiment prediction.

The system also incorporates key functionalities such as real-time web scraping, TF-IDF-based feature extraction, emotion detection, smart reply generation, and interactive dashboard visualization. These features collectively enhance the ability of businesses to understand customer opinions, identify product strengths and weaknesses, and make informed decisions.

The modular and scalable architecture of the system ensures flexibility and ease of integration with additional components in the future. Despite certain limitations, such as dependency on data quality and challenges in handling complex linguistic expressions, the system demonstrates strong potential for real-world applications in the e-commerce domain.

Overall, the proposed solution highlights the effectiveness of combining AI-driven techniques with practical system design to address the challenges of analyzing large-scale unstructured data. It provides a valuable tool for businesses to gain actionable insights from customer feedback and improve their services. With further advancements and enhancements, the system can evolve into a more comprehensive and intelligent platform for customer sentiment analysis.

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